Practice Quiz: Learning

Name: __ ***Date:*** __

Part A: Multiple Choice

1. Learning in Active Inference differs from perception because: A) Learning is faster B) Learning updates the model's parameters (long-term expectations), not just current beliefs C) Learning only happens consciously D) Learning does not involve prediction errors

2. The A matrix in a generative model represents: A) How actions affect the environment B) How hidden states cause observations (likelihood mapping) C) The agent's preferences D) The initial state of the system

3. Bayesian Model Reduction (BMR) is the process of: A) Adding more complexity to the model B) Evaluating whether parts of the model are useful and pruning those that aren't C) Forgetting everything and starting over D) Increasing the learning rate

4. As a skill becomes automatic, what happens to prediction errors? A) They increase because the skill becomes harder B) They decrease because the generative model becomes well-calibrated C) They stay the same D) They become random

5. Sleep is important for learning because: A) The brain is inactive during sleep, allowing memories to settle B) Slow-wave sleep consolidates parameters and REM sleep may prune model complexity C) Sleep has nothing to do with learning D) Dreams teach new information

6. The B matrix represents: A) How observations map to states B) How states change over time (transition probabilities) C) The agent's preferences D) The Markov Blanket structure

7. An experienced chess player considering fewer moves than a beginner is an example of: A) Laziness B) Structure learning — the expert's model has been pruned to focus on useful patterns C) Forgetting how to play D) Lower intelligence

Part B: Short Answer

1. You taste a new food that looks familiar but tastes very different from what you expected. Explain what happens in terms of prediction error and model updating. Which matrix (A, B, or D) is being updated?

2. Explain why cramming for an exam the night before is generally less effective than studying over several days with sleep in between. Use the concepts of parameter consolidation, structure learning, and sleep.

3. A toddler initially calls all four-legged animals "doggy." Over time, they learn to distinguish dogs, cats, horses, and cows. Describe this developmental process in terms of (a) initial model, (b) prediction errors that drive learning, and (c) the final, refined model.